Rama Steel Tubes Limited

IS 4923:2017

CML No. :- 7700047414

Work : Survey No. 151,Village - Umbare, Khopoli-Pali Road,Taluka - Khalapur, Dist. - Raigad – 410203. (MH)

Manufacturer's Test Certificate

T.C.No:------ 130123A | Customer :- SHREEJI STEEL TUBES | Invoice No. :- RSTL/2095

Date:-11/01/2023 Order No.: -

Production Description	SL NO	Size	Batch No /Heat No	NO OF PICES	Diamention of Test Piece			Max Load	Tensile Strength	Elongation %	Yield load	Yield stress	Straightness	Twist	Concavity	Convexity	Squareness	Radius of Corner	Remark
					Width MM	ThicknessMM	AreaMM	NEWTON	Mpa		NEWTON	Mpa							
BLACK STEEL TUBES & PIPES	1	32X32X2.00X6000mm	110123A	243	20.00	2.01	40.20	16790	418	30	11750	292	3	0.22	0.26	0.22	91	4.0	Ok
	2	32X32X2.50X6000mm	110123A	260	20.00	2.52	50.40	19850	394	32	13890	276	2	0.2	0.18	0.28	91	4.5	Ok
	3	32X32X3.00X6000mm	110123A	284	20.00	3.01	60.20	23430	389	35	16400	272	3	0.24	0.24	0.26	91	5.0	Ok

Note : All the visual finish & Dimensional Characteristics like OD Length, Thickness, weight ,kg. ,Meter are exactly copying with the specified limit & tolerance As per IS:4923

CHEMICALS ANALYSIS

C	S	MN	P	Result
0.052	0.008	0.36	0.007	OK

RAMA STEEL TUBES LTD. Umbre, Raigad

Authorised Signatory

S MAHIPAL STEEL SMS BHOSARI, PUNE-411 026

MURBAD, INDIA

TEST CERTIFICATE FOR STEEL TUBES

Format No.: QA-F-18
Rev.no. : 00
Date : 10.10.2021

APL APOLLO TUBES LIMITED UNIT-III

Works-Vill-Kudavali Plot No-M-1, Murbad, 421401, Thane (MS)

IS: 4923

CM/L-3850769

TEST CERTIFICATE NO :	184923/210/IS0750
DATE :	05-05-23
INVOICE NO	2005005272
INVOICE DATE	05-05-23

We certify that the material described below fully conformed to IS 4923 : 2017. Chemical composition and Mechanical properties of the product, as tested in tested in accordance with the scheme of testing and inspection contained in the BIS certification marks license no. CM/L-3850769 are as indicated below against each order no.

TEST RESULT

Size	Grade/ Type	HEAT NO./ BATCH NO.	Pcs.	TOTAL Wt (MT)	CHEMICAL ANALYSIS (% Max)				MECHANICAL PROPERTIES (Min)			GALVANIZING (Min)	
					C	Mn	P	S	Tensile Strength (Mpa)	Yeild Strenth (Mpa)	% El	Weight (g/m2)	Uniformity
Requirements					**0.12**	**0.60**	**0.04**	**0.04**	**330**	**210**	20	N/A	N/A
150 X 150 X 6.00 mm	BLK SHS (YST-210)	**20303978 / 8201121A**	72	9.57	0.048	0.118	0.008	0.007	385	305	31	–	–
60 X 60 X 5.00 mm	BLK SHS (YST-210)	**20104265 / 7201202A**	108	5.1	0.05	0.215	0.012	0.006	367	299	38	–	–
80 X 40 X 5.00 mm	BLK RHS (YST-210)	**20204579 / 2201126B**	75	2.42	0.04	0.104	0.006	0.005	403	304	32	–	–
150 X 150 X 8.00 mm	BLK SHS (YST-210)	**20204445 / 8201121A**	36	7.53	0.037	0.145	0.008	0.005	398	309	33	–	–

It is certified that Material supplied conforms to standard dimensions and mass tolerances.

AUTHORISED SIGNATORY

S MAHIPAL STEEL BHOSARI, PUNE

PREPARED BY

FOR APL APOLLO TUBES LIMITED

APL APOLLO TUBES LIMITED UNIT-III

Works-Vill-Kudavali Plot No-M-1, Murbad, 421401, Thane (MS)

TEST CERTIFICATE FOR STEEL TUBES

IS: 4923

CM/L-3850769

TEST CERTIFICATE NO :	IS4923/210/IS01073
DATE :	13-06-23
INVOICE NO	1805823293
INVOICE DATE	13-06-23

M/s Apl Apollo Tubes Ltd Pune

We certify that the material described below fully conformed to IS 4923 : 2017. Chemical composition and Mechanical properties of the product, as tested in tested in accordance with the scheme of testing and inspection contained in the BIS certification marks license no. CM/L-3850769 are as indicated below against each order no.

TEST RESULT

Size	Grade/ Type	HEAT NO./ BATCH NO.	PCS.	Actual Wt (MT)	CHEMICAL ANALYSIS (% Max)				MECHANICAL PROPERTIES (Min)			GALVANIZING (Min)	
					C	Mn	P	S	Tensile Strength (Mpa)	Yeild Strenth (Mpa)	%El	Weight (g/m2)	Uniformity
Requirements					0.12	0.60	0.04	0.04	330	210	20	N/A	N/A
145 X 82 X 5.00 mm	BLK RHS (YST-210)	19100274/ 8190210B	10	0.96	0.037	0.127	0.012	0.005	400	307	34	–	–
100 X 100 X 5.00 mm	BLK SHS (YST-210)	29890J/ 6190131B	14	0.74	0.059	0.391	0.011	0.0059	371	270	36	–	–
122 X 61 X 6.00 mm	BLK RHS (YST-210)	29903J/ 6190206B	9	0.41	0.059	0.391	0.011	0.0059	375	274	38	–	–
100 X 50 X 5.00 mm	BLK RHS (YST-210)	L03576/ 6180916A	13	0.78	0.071	0.38	0.007	0.007	396	300	32	–	–

It is certified that Material supplied conforms to standard dimensions and mass tolerances.

S MAHIPAL STEEL SMS PUNE-411 026

PREPARED BY

FOR APL APOLLO TUBES LIMITED

Balajee Structurals (India) Private Limited

Tatibandh, Raipur - 493221, Chhattisgarh, India

Test Certificate for Hot Rolled Medium & High Tensile Structural Steel as per IS 2062:2011

CM/L-8968720

Test Certificate No. : BSIPL/QC/2023-24/968 Date: 03-JAN-2024

To,

613

We certify that the material described below fully conforms to IS 2062:2011 Chemical Composition & Mechanical Properties of the product as tested in accordance with the scheme of Testing & Inspection contained with the documents for ISI Certification Marks Licence No. CM/L-8968720 are as indicated below against each order no.

(PLEASE REFER TO IS 2062:2011 FOR DETAILS OF SPECIFICATION REQUIRMENTS)

TEST RESULTS

Order no. & Date	Nominal Size in (mm)	Lot No.	QTY in (MT)	Chemical Analysis								Mechanical Properties						Grade	Remarks
				C %	Mn %	P %	S %	Si %	N %	Micro Alloying Ele. %	CE %	Yield Stress N/mm²	Tensile Strength N/mm²	Elon. % Min	Bend Test 180° 2t	Impact Test	Crack ability test		
				Max	Max	Max	Max	Max	Max		Max			23					
				0.23	1.50	0.045	0.045	0.40	0.012		0.42	Min 250	Min 410						
6271 DT. 30-DEC-2023	ISMC 150X75	22303-C	2.050	0.17	0.65	0.038	0.034	0.19	....	....	0.28	287	494	29	Ok	....	....	E-250A	Ok
	ISMB 150X75	22294-B	4.030	0.14	0.51	0.029	0.026	0.15	....	....	0.22	292	511	32	Ok	....	....	E-250A	Ok
	ISMB 200X100	22304-B	6.050	0.19	0.70	0.039	0.036	0.21	....	....	0.30	307	505	32	Ok	....	....	E-250A	Ok
	ISA 100X100X10	22213-A	2.170	0.17	0.63	0.035	0.030	0.18	....	....	0.27	286	488	28	Ok	....	....	E-250A	Ok
Total			14.300																

*For Copper-Bearing Quality

Micro alloying element present should be indicated.

The material supplied conforms to the standard rolling and tolerances.

PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys
C.I.S.O
VIDC., OUR DC/INV. No.
For YASH ENTERPRISES

Shiping/Invoice/Challan No.: 4139
Invoice Date: 03-JAN-2024
Wagon/ Truck No.: MH12 LT 7281

For, Balajee Structurals (India) Pvt. Ltd.

BALAJEE STRUCTURALS (INDIA) PVT. LTD. RAIPUR

Quality Control Incharge

RASHTRIYA ISPAT NIGAM LIMITED

VISAKHAPATNAM STEEL PLANT
QUALITY ASSURANCE & TECHNOLOGY DEVELOPMENT DEPARTMENT
VISAKHAPATNAM -530031 (A.P)

IS 2062

CM/L-2338754

TEST CERTIFICATE

DETAILS	CUSTOMER/DESTINATION	WAGON/TRUCK/TRAILOR No.	R.R. WEIGHT (TONS)
T.C.NO: QMS/R/QAD/RM/15/M/2023-24/9000/00136406 DATE: 27.12.2023 INDENT NO: 231344		SWR21150641295	64.700

WE CERTIFY THAT THE MATERIAL DESCRIBED BELOW FULLY CONFORMS TO IS 2062 - 2011 CHEMICAL COMPOSITION OF THE PRODUCT, AS TESTED IN ACCORDANCE WITH THE SCHEME OF TESTING AND INSPECTION CONTAINED IN THE BIS CERTIFICATION MARKS LICENSE NUMBER CM/L-2338754 ARE AS INDICATED AGAINST EACH ORDER NUMBER
(PLEASE REFER TO IS 2062 - 2011 FOR DETAILS OF SPECIFICATION REQUIREMENTS)

HEAT / LOT NUMBER	NOMINAL SIZE (mm)	NO. OF COILS/ BUNDLES/ BILLETS/ BLOOMS	CHEMICAL COMPOSITION									MECHANICAL PROPERTIES						COLOR	GRADE/ DESIGNATION
			C	Mn	P	S	Si	Al	Cr	V	Ce	YS	UTS	%EL	RA	BEND TEST	IMPACT TEST(J)		
			%	%	%	%	%	%	%	%	%	N/mm^2	N/mm^2	%	%				
311634	100X50X5	2	0.160	0.800	0.028	0.035	0.100	0.010	0.015	---	0.296	317.000	486.000	31.000	0.000	OK	--	YELLOW	IS2062 E250BR
311792	100X50X5	4	0.200	0.850	0.019	0.038	0.090	0.015	0.016	---	0.345	309.000	453.000	32.000	0.000	OK	--	YELLOW	IS2062 E250BR
311807	100X50X5	1	0.170	0.850	0.026	0.021	0.040	0.023	0.005	---	0.313	329.000	492.000	32.000	0.000	OK	--	YELLOW	IS2062 E250BR

STANDARD SPECIFICATION:

CHEMICAL	C	Mn	P	S	Si	Al	Cr	Cu	Ce	V	TI	B
	0.030 - 0.220	0.100 - 1.500	0.005 - 0.045	0.005 - 0.045	<= 0.400	0.002 - 0.120	<= 0.050	NA	0.050 - 0.470	---.	---	NA

MECHANICAL	YS	UTS	%EL	RA
	250 - 350	410 - 550	>= 23	>= 25

THE MATERIAL SUPPLIED CONFORMS TO THE STANDARD ROLLING AND MASS TOLERANCES.

AVERAGE WEIGHT OF EACH BUNDLE/COIL/BILLET/BLOOM IS 9.243 TONS APPROXIMATELY
MANUFACUTRING ROUTE: BOF-LF/IRUT-CCM-CHANNEL
REDUCTION RATIO:

CERTIFIED THAT
PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys,
REC. NO. DATED
VIDC., OUR DC/INV. No.....
Our Vas... Enterprises

FOR (QA & TD)
VISAKHAPATNAM STEEL PLANT
R.I.N.L

SHREE HANUMAN LOHA PVT. LTD.

251/252 'C' URLA INDUSTRIAL AREA, RAIPUR (C.G.)

TEST CERTIFICATE FOR STEEL GENERAL STRUCTURAL PURPOSE

Test Certificate no: SHLL/RYP/22-23/ 3954

DATE : 27.12.2023

This is to certify that material supplid to you vide our invoice no. 3954

BUYER

M/S. **YASH ENTERPRISES**
PUNE

CONSIGNEE

M/S **YASH ENTERPRISES**
PUNE

We certify that materials described below fullyconfirms to.IS.2062-2011Grade-E250

TEST RESULTS

ITEM Nominal SIZE NO.	COST LOT	QUANTITY MT	C%	S%	P%	MN%	CU %	SI%	CE%	Y.S.N./MM	Tensile Strenth N/MM	Elongati ON%	Bend test	GRADE	REMARK
M.S.Angle 50x50x5		2.200	0.17	0.035	0.045	0.57	...	0.23	0.265	262	440	29.00	OK	A	
M.S.Angle 75X75X8		1.600	0.19	0.035	0.044	0.55	...	0.22	0.282	285	417	30.00	OK	A	
M.S.Angle 100X100X8		1.000	0.19	0.033	0.046	0.62	...	0.21	0.258	305	440	30.00	OK	A	
M.S.Angle 100X100X12		2.140	0.19	0.033	0.046	0.62	...	0.21	0.258	305	440	30.00	OK	A	
CHANNEL 100X50		10.000	0.19	0.033	0.043	0.55	...	0.22	0.282	279	498.00	30.00	OK	A	
CHANNEL 75x40		8.000	0.18	0.034	0.042	0.56		0.24	0.270	280	476.00	29.00	OK	A	
TOTAL QUANTITY		24.940													

Four copper beaming quality

Our Brand Name Is Balajee

The material supplied conforms to the standard rolling and weight tolerances

WAGON / TRUCK NO :- MH14HG7767

FOR; SHREE HANUMAN LOHA PVT. LTD.
Quality Control Incharge

CERTIFIED THA
PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys
C-75
IDC., OUR DC/INV. No.
YASH ENTERPRISES

SHREE HANUMAN LOHA PVT. LTD.

251/252 'C' URLA INDUSTRIAL AREA, RAIPUR (C.G.)

TEST CERTIFICATE FOR STEEL GENERAL STRUCTURAL PURPOSE

Test Certificate no: SHLPL/RYP/23-24/ 3437 DATE : 24-11-2023

This is to certify that material supplid to you vide our invoice no. 3437

BUYER
M/S **YASH ENTERPRISES**
PUNE

CONSIGNEE
M/S **YASH ENTERPRISES**
PUNE

We certify that materials described below fullyconfirms to.IS.2062-2011Grade-E250

TEST RESULTS

ITEM Nominal SIZE NO.	COST LOT	QUANTITY MT	C%	S%	P%	MN%	CU %	SI%	CE%	Y.S.N./MM	Tensile Strenth N/MM	Elongati ON%	Bend test	GRADE	REMARK
M.S.Angle 50x50x5		1.000	0.17	0.035	0.045	0.57	...	0.23	0.265	262	440	29.00	OK	A	
M.S.Angle 100X100X8		2.100	0.19	0.033	0.046	0.62	...	0.21	0.258	305	440	30.00	OK	A	
M.S.Angle 100X100X10		2.100	0.18	0.033	0.047	0.58	...	0.21	0.260	286	457	31.00	OK	A	
CHANNEL 100X50		8.010	0.19	0.033	0.043	0.55	...	0.22	0.282	279	498.00	30.00	OK	A	
TOTAL QUANTITY		13.210													

Four copper bearing quality
Our Brand Name Is Balajee
The material supplied conforms to the standard rolling and weight tolerances

WAGON / TRUCK NO :- **MH16CD-9076**

FOR;
SHREE HANUMAN LOHA PVT. LTD.
Quality Control Incharge

CERTIFIED THA
PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys
A 100 X 10
VIDC., OUR DC/INV. No.....
YASH ENTERPRISES

IS 4923:2017

CML No. :- 7700047414

Rama Steel Tubes Limited

Work : Survey No. 151,Village - Umbare, Khopoli-Pali Road,Taluka - Khalapur, Dist. - Raigad – 410203. (MH)

Manufacturer's Test Certificate

T.C.No:------ 040124

Invoice No. :- RSTL/1584

Date:-03/01/2024 Order No.: -

Production Description	SL NO	Size	Batch No / Heat No	NO OF PICES	Diamention of Test Piece			Max Load	Tensile Strength	Elongation %	Yield load	Yield stress	Straightness	Twist	Concavity	Convexity	Squareness	Radius of Corner	Remark
					Width MM	ThicknessMM	AreaMM	NEWTON	Mpa		NEWTON	Mpa							
BLACK STEEL TUBES & PIPES	1	60X60X4.00X6000MM	170223A	70	20	4.01	80.2	30000	374	30	21000	262	2.2	0.28	0.22	0.24	90	10.5	Ok
	2	80X80X4.00X6000MM	170223A	19	22	4.01	88.22	32000	363	29	26000	295	2.2	0.28	0.22	0.24	91	11	Ok
	3	80X80X5.00X6000MM	101023B	34	20	5.02	100.4	37000	369	32	26500	264	2.8	0.28	0.22	0.24	91	13	Ok
	4	122X61X5.00X6000MM	121123B	15	21	5.02	105.42	40000	379	29	28500	270	2.5	0.28	0.22	0.24	91	14	Ok
	5	150X150X5.00X6000MM	221223B	15	20	5.01	100.2	38000	379	30	26500	264	2.5	0.28	0.22	0.24	91	14.5	Ok
	6	100X100X5.00X6000MM	251223B	18	20	5.02	100.4	38800	386	30	27000	269	3	0.28	0.22	0.24	90	13	Ok
	7	100X100X6.00X6000MM	170223A	28	20	5.95	119	44000	370	34	31500	265	3.2	0.28	0.22	0.24	91	15	Ok
	8	145X82X5.00X6000MM	170223A	21	20.5	5.02	102.91	40500	394	30	30000	292	3.25	0.28	0.22	0.24	91	16.5	Ok

Note : All the visual finish & Dimensional Characteristics like OD Length, Thickness, weight ,kg. ,Meter are exactly copying with the specified limit & tolerance As per IS:4923

CHEMICALS ANALYSIS				
C	S	MN	P	Result
0.052	0.011	0.39	0.006	OK

CERTIFIED THAT PERTAINING TO THIS CERTIFICATE HAVE BEEN SOLD TO M/s. Technosys REC 145X82X5 VIDC., OUR DC/INV. No.....

For YASH ENTERPRISES

RAMA STEEL TUBES LTD. Umbre, Raigad

Authorised Signatory

APLAPOLLO — MURBAD, INDIA

TEST CERTIFICATE FOR STEEL TUBES

Format No.: QA-F-18
Rev.no. 00
Date : 10.10.2017

APL APOLLO TUBES LIMITED UNIT-III

Works-Vill-Kudavali Plot No-M-1, Murbad, 421401, Thane (MS)

IS: 4923

CM/L-3850769

TEST CERTIFICATE NO :	**IS4923/210/IS0859**
DATE :	27.10.2023
INVOICE NO	**2005821605**
INVOICE DATE	27.10.2023

We certify that the material described below fully conformed to IS 4923 : 2017. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification marks license no. CM/L-3850769 are as indicated below against each order no.

TEST RESULT

Size	Grade/ Type	HEAT NO./ BATCH NO.	Pcs.	TOTAL Wt (MT)	CHEMICAL ANALYSIS (% Max)				MECHANICAL PROPERTIES (Min)			GALVANIZING (Min)	
					C	Mn	P	S	Tensile Strength (Mpa)	Yeild Strenth (Mpa)	%El	Weight (g/m2)	Uniformity
Requirements					**0.12**	**0.60**	**0.04**	**0.04**	**330**	**210**	**20**	**N/A**	**N/A**
30 X 30 X 3.00 mm	BLK SHS (YST-210)	20304545 / 4210111A	820	13.120	0.046	0.152	0.029	0.007	392	210	32	–	–

It is certified that Material supplied conforms to standard dimensions and mass tolerances.

Original

PREPARED BY

CERTIFIED THAT PERTAINING TO THIS CERTIFICATE HAVE BEEN SOLD TO M/s. Technosys SQ 30x3 VIDE OUR DC/INV No.

For VASU ENTERPRISES

AUTHORISED SIGNATORY

FOR APL APOLLO TUBES LIMITED

JSW STEEL LIMITED

Dolvi Works, Taluka - Pen, District - Raigad, Maharashtra - 402[illegible], India
Regd. Office : JSW Centre, Bandra Kurla Complex, Bandra (East) , Mumbai - 400 051
CIN NO. : L27102MH1994PLC152925

IS 3502
CML-2878786

TEST CERTIFICATE FOR STEEL CHEQUERED PLATES

ertificate No.: 7106334315

Date : 20.12.2023
Product : Hot Rolled Coils
SO No. : 402140661
SO Date : 16.12.2023

We certified that the material described below fully conforms to IS 3502:2009. Chemical composition and Mechanical properties of the product, as tested in accordance with the Scheme of Testing and Inspection contained in the BIS Certification Marks Licence No.CM/L-2878786, are as indicated below against each order No.
(PLEASE REFER TO IS 3502:2009 FOR DETAILS OF SPECIFICATION REQUIREMENTS)

Specification : IS 3502:2009 MTL_IS2062_E250BR — Chemical Composition

		C %	Mn %	S %	P %	Si %	Al %	N ppm	B ppm	Nb %	V %	Ti %	Cr %	Mo %	Ni %	Cu %	MAE %	C Eq %	Killing
Specification Requirements	Min					0.030	0.010										0		
	Max	0.220	1.500	0.045	0.045	0.400		120									0		

Test Results

Coil No. / Packet No.	Nominal Size (mm) T X W X L	Pcs	Qty. MT																	
2320554601	5.00 x 1250 x C		20.535	0.0490	0.700	0.0040	0.015	0.075	0.036	51		0.016						0.005	0.016	0.1660
Total weight in Metric Tonnes			20.535	Grand total of coils / packets							1									

Specification : IS 3502:2009 MTL_IS2062_E250BR — Mechanical Properties

		Tensile direction	YS MPa	UTS MPa	GL mm	El %	YS/UTS ratio	Bend direction	Bend dia., mm	Bend result	CVN Impact direction	CVN Impact temp., °C	CVN Impact avg energy J	Hardness HV_{10}	Hardness HRb	GS No.	IR	HER %	ECV mm	SET
Specification Requirements	Min	T	250.0	410.0		23.0		T												
	Max																			

Test Results

Coil No. / Packet No.	Nominal Size (mm) T X W X L	Pcs	Qty. MT																	
2320554601	5.00 x 1250 x C		20.535	T	418.00	488.00	5.65SR	29.00	0.857	T	2.0t	Ok								

ertify that the above mentioned products produced and supplied by JSW Steel Ltd, Dolvi works do not contain any radioactive element higher than the natural level. The product or packing material does not contain any hazardous substances as per
ms

oc No. :7106334315
No. :23DO2700268287
transport :Truck
No. :MH46AF6583

Process Route : CONARC-LHF-CSP.
Fully killed steel
Legends : CONARC = ConArc Furnance,LHF = Laddle Heating Furance, CSP =Compact Strip Production
T x W x L = Thickness x Width x Length
Chemical analysis = Laddle sample analysis, 1 MPa = 1N/mm2
GL = Gauge Length,YS = Yield Strength UTS = Ultimate Tensile Strength, El = Total elongation on standard GL;
CVN = Charpy V-notch, L = Longitudinal, T = Transverse °C = Degree Centrigrade, GS = ASTM Grain Size,
IR = Inclusion Rating, ECV = Erichsen Cupping Value, SET = Strainag Embrittlement Test,
MAE = Micro Alloying Elements, C Eq% = Carbon Equivalent= [C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15]
HER = Hole Expansion Ratio, RoHS = Restriction of Hazardous Substances
The material supplied conforms to the specified dimensions and tolerances

CERTIFIED THAT PERTAINING TO THIS CERTIFICATE HAVE BEEN SOLD TO M/s. Technosys
5mmx1250
VIDC., OUR DC/INV. No.
VASU ENTERPRISES

Pankaj Khasne
Deputy General Manager
Quality and System
For JSW Steel Ltd.

AM/NS INDIA

ArcelorMittal Nippon Steel India Limited

HAZIRA - 394270, DIST. SURAT, GUJARAT, INDIA
CIN:U27100GJ1976FLC013787

MILL TEST CERTIFICATE

FOR STEEL CHEQUERED PLATES

IS: 3502

Test Certificate No.	AMNS/TCH/20230128222829042464O	Test Certificate Date	29.01.2023
Sale Order No.	2120447138 / 20	Sales Order Date	24.01.2023
Equivalent Specification	IS_3502_2009	Product Type	HR-COIL
Specification	IS_3502_2009	Grade	SRCHP03
Invoice No.	GJ20221000180637		
Vehicle No.	DN09Q 9673		

WE CERTIFY THAT THE MATERIAL DESCRIBED BELOW FULLY CONFORMS TO ABOVE SPECIFICATION CHEMICAL COMPOSITION AND MECHENICAL PROPERTIES OF THE PRODUCT, AS TESTED IN ACCORDANCE WITH THE SCHEME OF TESTING AND INSPECTION CONTAINED IN THE BIS CERTIFICATION MARKS LICENCE NO. CM/L -7200048296 ARE AS INDICATED BELOW AGAINST EACH ORDER NO. ETC.

Heat Analysis (%) of P18179

Heat No	C	MN	S	P	SI	AL	Cu	N	MAE	CE
P18179	0.158	0.830	0.0080	0.010	0.155	0.034	0.012	0.0045	0.005	0.301

Mechanical Properties for Batches of P18179

Item No	Batch ID	Thk	Wid	Qty Net	YS	UTS	EL	BEND
20	LS5662	5.000	1500	25.520	308	450	26	OK
Total				25.520				

CERTIFIED THAT
PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys
5mm x 1500
VIDE., OUR DC/INV. No.....

for YASH ENTERPRISES

Note 1) Test Certificate conforms to EN 10204-3.1
Note 2) PROCESS ROUTE: HBI/DRI/BF/COREX->EAF/CONARC->LF->CCM->HSM/CSP and Steel is Fully Al-Si Killed
Note 3) The material supplied conforms to the standard rolling & weight tolerances.
Note 4) 1MPa=1N/mm2= 0.102Kgf/mm2.

Legends: Bend Diameter:3t, Bend Angle:180deg, Bend Direction:90deg, CE: Carbon Equivalent, EL: % Elongation on G.L = 5.65*SQRT(AREA), MAE: Nb+Ti+V, UTS: Ultimate Tensile Strength, YS: Yield Strength

Units of measurement: Thk/Wid/Len(MM), Qty Grs/Net(Quantity Gross/Net)(MT), EL(%), UTS(MPA), YS(MPA)

MANAGER (QUALITY ASSURANCE)
FOR ArcelorMittal Nippon Steel India Limited
AMNSIL/QAC/0003/F

**This document is electronically generated and hence does not require any signature.

PUSHPAK STEEL INDUSTRIES PVT. LTD.

GAT NO.119, DHANORE, ALANDI MARKAL ROAD, TAL. KHED, DIST. PUNE

TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL STEEL
TEST CERTIFICATE

TEST CERTIFICATE NO. : PSI / 01-24/2551

To,

YASH ENTERPRISES, BHOSARI , PUNE.

DATE : 09.01.2024

Consignee:-

We certify that the material described below fully conforms to IS 2062-2011 GR-E 250 A
Chemical composition and mechanical properties of the product as tested are given below.

TEST RESULT

Order No. Date	Article & Type (Nominal size) mm	Cast / Lot No.	Quantity	CHEMICAL ANALYSIS								MECHANICAL PROPERTIES						Grade
				C	Mn	Si	S	P	Cu *	Micro ** Alloying	CE	Yield Stress	Tensile Strength	Elongation	Bend Test Internal bend	Impact Test	Y Groove Crackability Test	
			MT	%	%	%	%	%	%	%	%	Mpa	Mpa	%	2t			
	CHANNEL MC 100X50	PSI/12-23/02	6.975	0.20	0.55	0.17	0.036	0.046	--	--	0.39	338.0	509.0	30.1	OK	--	--	E250A
												335.0	504.0	30.2	OK	--	--	E250A
	MS ANGLE 50X50X6	PSI/12-23/22G	2.010	0.21	0.52	0.22	0.033	0.044	--	--	0.36	333.0	488.0	30.0	OK	--	--	E250A
												331.0	487.0	30.5	OK	--	--	E250A
	MS ANGLE 65X65X6	PSI/12-23/11E	1.010	0.22	0.53	0.21	0.034	0.039	--	--	0.39	336.0	509.0	29.6	OK	--	--	E250A
												340.0	512.0	29.4	OK	--	--	E250A

The material conforms to the standard rolling and mass tolerances.

CERTIFIED THAT-
PERTAINING TO THIS CERTIFICATE
HAVE BEEN SOLD TO M/s. Technosys
C.I.O.O.
VIDC., OUR DC/INV. No.....
Format No. : PSI/F/QA/03 Rev.-00-

PUSHPAK STEEL INDUSTRIES PVT. LTD. Unit-I Gat No. 119 Dhanore Pune

Invoice No -AL- 3781/23-24 Date : 09.01.2024
Vehicle No MH 13 CU 9777

For YASH ENTERPRISES